

# Prospective evidence for conversational medical AI is hard, but non-negotiable

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Trust in clinical artificial intelligence (AI) cannot be benchmarked into existence. It must be earned through rigorous prospective studies in real-world clinical settings, where the hardest lessons often concern the humans and systems around the AI, not the technology itself.

Large language models (LLMs) have shown impressive performance in clinical reasoning over recent years, consistently meeting or exceeding that of practicing physicians on standardized benchmarks<sup>1,2</sup>. These gains have not happened in a vacuum; as performance improves, so has benchmark complexity, evolving from static multiple-choice question banks to multi-turn diagnostic and disease management conversations; from text to multimodal reasoning; and from general medicine to subspecialty domains<sup>3–5</sup>. These changes reflect genuine improvement in model performance and the science of evaluation over a short period of time.

However, with this explosion in evaluation, a recent systematic review concluded with the sobering observation that “rigorous, patient-centered evidence remains scarce”<sup>6</sup>. Patient-facing clinical AI has unique evaluation challenges that stretch far beyond benchmarking. Understanding these intricacies requires more than technical accuracy; it also demands evaluation of real-world performance in a safe and controlled environment. Although conventional benchmarks, simulated encounters and retrospective studies provide key insights, they cannot fully establish how these systems are used by real patients, which is crucial to inform patient-centered design<sup>7–9</sup>.

Patient-facing technologies need real-world research deployment to generate evidence. Clinicians also need evidence before changing their workflows. Technological capabilities alone are necessary, but not sufficient to build trust and clinical adoption.

We describe lessons learnt from our research with the Articulate Medical Intelligence Explorer (AMIE), a research AI system for clinical reasoning and conversations, which was initially developed and assessed using *in silico* simulations and structured clinical assessments with patient actors, and then moved beyond simulated settings towards two progressively larger prospective clinical studies. We argue that prospective, clinical-grade evidence generation, despite being more difficult than benchmarking studies, should become a standard for conversational AI in health and medicine.

## Two prospective studies of AMIE in real-world clinical settings

These two Institutional Review Board (IRB)-approved studies represent a progressive journey from single-site feasibility to a nationwide randomized study (Fig. 1).

**Study 1.** Study 1 was a prospective, single-arm feasibility study conducted collaboratively by researchers at Google and Beth Israel Deaconess Medical Center (BIDMC) in Boston, Massachusetts. To our knowledge, this was the first prospective pre-registered (ClinicalTrials.gov ID [NCT06911398](https://clinicaltrials.gov/ct2/show/study?term=NCT06911398)) evaluation of a patient-facing LLM-based conversational AI in a real-world clinical setting<sup>10</sup>. In total, 100 adult patients completed a text-based conversation with the AMIE before their urgent care appointment, under real-time physician safety supervision. The study assessed conversational safety, conversation quality, clinical reasoning performance (via blinded comparison with primary care providers (PCPs)), and patient and clinician experience through surveys and qualitative interviews.

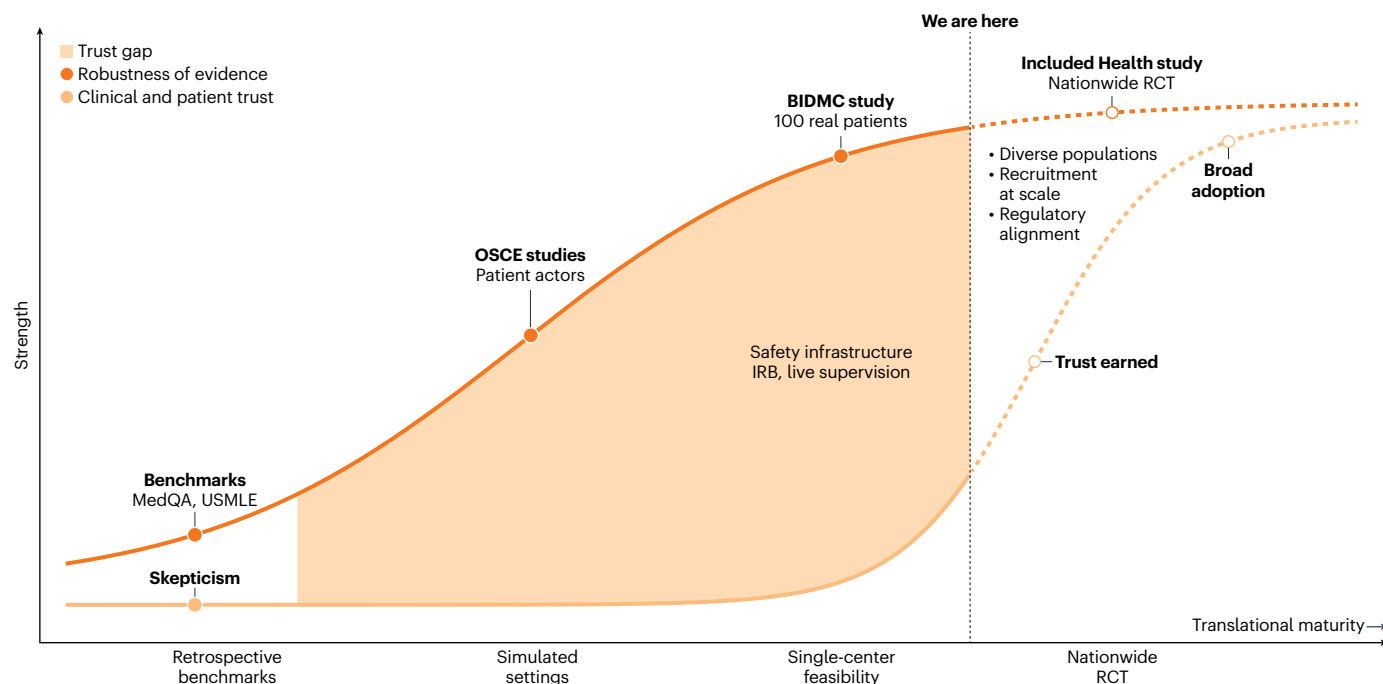
**Study 2.** Study 2 is an ongoing, pre-registered, nationwide randomized study in the USA, conducted collaboratively by Google and the virtual care provider Included Health. Using controlled comparisons, the study assesses AI in real-world virtual care settings across common complaints, including respiratory, cardiovascular, musculoskeletal, genitourinary, reproductive and gastrointestinal domains.

From our experiences designing and executing prospective studies, we have identified four themes: from the initial clinical question (can this be done safely?) to the sociotechnical foundation that underlies each step of execution (what does it take to generate this evidence and scale AI use?).

## Demonstrating safety with the right infrastructure

At BIDMC, across all 100 AMIE–patient encounters, physician safety supervisors did not need to intervene in any consultation based on predefined safety criteria<sup>10</sup>. Studying conversational safety of AI in this setting required careful coordination; researchers worked with quality and safety teams from the BIDMC and the IRB to develop a deliberately conservative approach. Dedicated remote physicians monitored every interaction; high-risk presentations were excluded; and clinical red-teaming identified edge cases before enrollment began. Although live one-to-one supervision was essential in a first-in-human study, it may introduce Hawthorne effects and is unlikely to be scalable. Future research should investigate how oversight can be safely and progressively relaxed<sup>11</sup>.

From the beginning, the design of the system behavior was informed by cross-functional reviews across disciplines and institutions, often with difficult decisions. For example, how should the AMIE output diagnostic possibilities? We ultimately determined that presentation of a broad differential diagnosis was more consistent with patient expectations and clinician mental models during the diagnostic process as opposed to numerical likelihood estimates that are difficult to validate and challenging to convey the correct message to the participant. Although these are seemingly small choices



**Fig. 1 | Building trust in conversational and reasoning clinical AI.** Progression of AMIE's evidence base from retrospective benchmarks and simulated settings with patient actors, following Objective Structured Clinical Examination (OSCE) assessment formats, to real-world prospective clinical assessment including

a single-center feasibility study with BIDMC and a nationwide randomized controlled study (RCT) with Included Health, involving IRB approval and safety infrastructure such as live clinical supervision of AI–patient interactions.

in study design, they were developed through deliberate discussions with several stakeholders.

## AI interventions reshape the entire clinical encounter

Even when introduced in a study setting, conversational AI restructures the clinical encounter in ways that can be hard to predict or measure. At BIDMC, PCPs reported that AMIE shifted consultations “from data gathering to data verification”; patients arrived organized with coherent narratives, enabling more collaborative conversations and shared decision-making. Moreover, PCPs reported that they found preparing the urgent care appointment with AMIE either ‘very’ or ‘somewhat’ helpful in 75% of cases. We tried to select patients with a range of technological literacy; notably, 54% of patients did not score highly on measures of technological literacy<sup>10</sup>.

We were surprised that some patients chose to share remarkably detailed clinical information, including full medication lists, through multi-turn interactions that went well beyond what standard intake forms capture. This suggests future opportunities for patient and medication reviews that may be more comprehensive than is typically possible within routine clinical time constraints. Although this may hint at a patient cohort that is highly motivated to participate, one participant explained their in-depth engagement with the AI conversations as a “judge-free zone” for disclosing symptoms without face-to-face pressure.

However, these positive effects only materialized when AI output reached the clinician. At BIDMC, 27% of PCPs did not review the AMIE transcript before the visit, because it arrived asynchronously from their preparation workflow rather than through the electronic health records. When clinicians did not incorporate the output, patients were

frustrated by the need to repeat themselves. This ‘handoff failure’ made the interaction feel redundant. The study highlighted the promise of triadic care, with AI enhancing the patient–doctor relationship, but also the workflow barriers to realizing it.

In the Included Health study, in which the handoff is more streamlined in the form of a brief digital transition (since both the AI interaction and PCP care are virtual and sequential), both patient and clinician experiences are notably positive. However, virtual care introduces its own challenges. In the run-in phase, some patients left the encounter before seeing the clinician. This may be because the research study is attracting less motivated users, or because patients can more easily ‘walk away’ in a virtual setting before seeing the clinician. Further evaluation is needed to confirm usage and dropout rates and to reveal the promise of triadic care models.

## Trust is earned through experience, not just performance

High performance on clinical reasoning in the real world does not necessarily translate to clinical acceptability. At BIDMC, although AMIE performed comparably to PCPs on blinded metrics, fewer than 50% of patients felt the system appeared “honest and trustworthy” or would maintain confidentiality despite clear communication about data privacy safeguards during the informed consent step<sup>10</sup>. Real-world self-reported trust was lower than in simulated settings, which is another aspect that benchmarks and simulated patients cannot predict.

Our results showed that trust is not static and can be earned or lost based on interactions with the AI system. In the context of BIDMC, patients’ attitudes towards AI improved significantly after a single AMIE interaction and remained increased after the subsequent

PCP visit<sup>10</sup>. These trust profiles probably vary across different care settings: patients at the BIDMC, who were seeing their own physicians, valued AMIE as “empathetic and human-like”, whereas some patients in the Included Health study valued pragmatism, focusing on privacy, transparency and cost savings rather than empathy<sup>10</sup>.

Trust is not a fixed property of a model and cannot be measured as a benchmark. It is earned incrementally through direct human experience with the AI agent and rigorous scientific analysis of the study results. Only prospective studies can observe this dynamic.

## The infrastructure of evidence

Perhaps the most under-discussed challenge in translational AI is the logistical burden of generating rigorous prospective evidence, what might be termed the ‘infrastructure of evidence’.

**Recruitment and selection bias.** At BIDMC, recruitment was primarily manual via phone calls from study staff based on eligibility criteria (adult patients with an existing single-complaint urgent care appointment). This was manageable for 100 participants, but not practical to generate population-level evidence at scale. The Included Health study recruited from several sources including the Included Health platform and the Google Health app, each bringing different participant populations and expectations that must be aligned within a standardized study protocol.

Across both studies, comfort with AI technology seems to drive participation in ways that need to be reported during data analysis. Future studies may need to oversample patients with less comfort with AI and lower health literacy.

**Methodological design.** During independent expert review of AI and PCP outputs, blinded clinical assessment requires navigating a trade-off: maintaining fidelity to the original clinical output while achieving blinding that reduces bias, whether positive or negative, about provenance (AI versus PCP). In the BIDMC study, this involved a generic LLM-based transform step with a design that was iterative, informed and reviewed by the clinical team to ensure no clinically significant changes are introduced by the blinding transformation<sup>10</sup>. Our studies benefited from having a predefined run-in phase, prior to locking in a statistical analysis plan, to refine such procedures in data collection and analysis.

**AI infrastructure.** Mid-study at BIDMC, enrollment was paused to switch model backends owing to technical issues with the research infrastructure over the course of the study. This is a reality unique to modern AI clinical trials, in which ensuring the stability of experimental AI architectures may require substantial infrastructure effort or computational capacity. For studies conducted with commercial model APIs, an additional consideration is ensuring that a given model remains static throughout; for example, by pointing to a specific versioned endpoint guaranteed by the commercial provider to be static.

**Research partnerships require deliberate design.** Prospective AI studies demand deep collaboration between AI developers and clinical partners. How does collaborative study design work across organizations with different incentives? What is the formal role of each institution and team? What types of data flow between institutions, and how can we safeguard privacy and security? How should the possibility of clinician opt-out be addressed? How are adverse events monitored for and tackled? How much autonomy is granted to the AI system? Who can

override its recommendations, and what human oversight is expected? And if harm occurs, who is held accountable?

These questions must be addressed before enrollment begins. Clinical trial pre-registration should be standard practice, ensuring scientific rigor from the outset<sup>10,12</sup>.

## The road ahead

The field of conversational medical AI has built an extraordinary base of benchmark evidence. What is scarce, and what clinical adoption requires more of, is a corresponding base of prospective, real-world evidence demonstrating safety, efficacy, trust, and workflow integration.

For patient-facing conversational AI, this evidence should come from research studies that include real patients with real health concerns. These research studies should embed AI within actual clinical workflows rather than standalone interfaces; measure trust and experience longitudinally rather than at a single time point; and test across diverse settings, populations, and care contexts. While randomized controlled trials of patient-facing conversational AI are the next evidentiary frontier, there are scenarios that even controlled trials may be unable to capture fully: patients who fall outside typical inclusion criteria; uncooperative or adversarial interactions; and the risk of clinicians developing overreliance over extended periods of time. These are the realities of clinical practice, and they will require ongoing monitoring well beyond the boundaries of any single study<sup>13,14</sup>.

Early studies with self-selected, technology-comfortable participants are a necessary first step, but not the final destination. The involvement of diverse, representative populations, including those least served by current healthcare systems, is important. The AMIE research deployment journey, from 100 patients at a single academic center to a nationwide randomized trial, is just a single journey on this new ‘benchmark to bedside’, but many more are needed to advance the mission of improving access to care, reducing physician burden, and empowering patients<sup>15</sup>.

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